Phase 2: Data Design & Analysis

Manufacturing Downtime & Line Productivity Project — Tool: Microsoft Power BI — Deadline: November 6

1. Understanding the Dataset & Data Dictionary

Our dataset tracks production batches and machine downtime for a beverage bottling line from **August 29, 2024** to **June 30, 2025** (**286 active production days**). It contains **3,000 production batches** (IDs 422111 to 425110) organized across **4 core worksheets** that feed into our Power BI data model.

1.1 Summary of Workbook Sheets

Sheet Name	Role in Power BI	Size	What It Contains & How We Use It
Line productivity	Fact Table (**ProductionBatch**)	3,000 × 10	Main log of every batch: date, product, batch ID, operator, start/end time, and bottle counts (produced, target, good, and rejected).
Line downtime	Fact Table (**BatchDowntime**)	3,000 × 13	Records downtime minutes across 12 factor columns (**1** to **12**). We will unpivot this table in Power Query so it links cleanly in the model.
Products	Lookup Table (**Product**)	6 × 4	Reference list of the 6 drinks with their flavor, bottle size (**600 ml** or **2 L**), and minimum standard batch time (**60** or **98** mins).
Downtime factors	Lookup Table (**DowntimeFactor**)	12 × 3	Reference list of the 12 downtime reasons, including description and whether the issue is caused by operator error (**Yes** or **No**).

1.2 Data Dictionary

Table Name	Column Name	Power BI Type	Key / Role	Description & Actual Range in Dataset
ProductionBatch	**Batch**	Whole Number	Primary Key	Unique batch number from **422111** to **425110** (3,000 unique batches, no duplicates).
ProductionBatch	**Date**	Date	Foreign Key	Production date from **2024-08-29** to **2025-06-30** (286 active days, ~10.5 batches/day).
ProductionBatch	**Product** (**ProductID**)	Text	Foreign Key	Links to **Product** table. Values: **CO-600, LE-600, OR-600, CO-2L, RB-600, DC-600**.

Table Name	Column Name	Power BI Type	Key / Role	Description & Actual Range in Dataset
ProductionBatch	**Operator (OperatorID)**	Whole Number	Foreign Key	Operator name in raw Excel (**Charlie, Dee, Mac, Dennis**); replaced by **OperatorID** (1–4) in our model.
ProductionBatch	**Start Time / End Time**	Time	Attribute	Batch start and finish times (**HH:MM:SS**). Includes 64 overnight batches that cross midnight.
ProductionBatch	**Units Produced**	Whole Number	Fact Metric	Total bottles produced per batch. Range: **554 – 1,993**; Total: **4,185,326** units.
ProductionBatch	**Target Quantity**	Whole Number	Fact Metric	Planned target per batch. Range: **650 – 1,800**; Total: **4,198,389** units.
ProductionBatch	**Good Units**	Whole Number	Fact Metric	Bottles passing quality checks. Range: **543 – 1,967**; Total: **4,089,359** units (**97.71%**).
ProductionBatch	**Rejected Units**	Whole Number	Fact Metric	Defective bottles rejected. Range: **1 – 124**; Total: **95,967** units (**2.29%** rejection rate).
Product	**ProductID, Flavor, Size**	Text	PK (**ProductID**)	Product details: 5 flavors (**Cola, Orange, Lemon lime, Diet Cola, Root Berry**) and 2 sizes (**600 ml, 2 L**).
Product	**Min batch time**	Whole Number	Attribute	Standard minimum time per batch: **60** mins for 600 ml bottles and **98** mins for 2 L bottles.
DowntimeFactor	**Factor, Description**	Whole # / Text	PK (**Factor**)	Factor code (**1** to **12**) and reason name (e.g., **Machine adjustment, Machine failure**).
DowntimeFactor	**Operator Error**	Text	Attribute	Shows if the stop is human error: **Yes** (Factors **2, 5, 6, 8, 10, 11**) or **No** (Factors **1, 3, 4, 7, 9, 12**).
Operator	**OperatorID, Name**	Whole # / Text	PK (**OperatorID**)	Lookup table created in Power Query for the 4 operators (**1: Charlie, 2: Dee, 3: Dennis, 4: Mac**).
BatchDowntime	**DowntimeRecordID, Batch, Factor, Duration**	Whole Number	PK & FKs	Created by unpivoting **Line downtime**. Contains **3,979** active downtime events (**5 – 75** mins; Total: **92,158** mins).

2. Data Inspection & Identified Problems

Before building our Power BI data model, we checked all 4 sheets for data quality issues. First, we confirmed that **there are 0 duplicate Batch IDs**, **0 missing values in Line productivity**, all Product and Factor codes match their lookup tables, and **Units Produced = Good Units + Rejected Units** is 100% accurate across all 3,000 rows. However, we found **5 key structural and data issues** that we need to fix in Power Query:

#	Problem Found	How Much Data Is Affected	Why It Matters in Power BI & How We Will Fix It
1	**Wide Table Layout in Line downtime**	**3,000** rows × **12** factor columns (**36,000** cells)	Power BI cannot link 12 separate factor columns to the **Downtime factors** table. We must use **Unpivot Other Columns** in Power Query to turn it into a 3-column table (**Batch, Factor, DowntimeDuration**).
2	**Blank Cells in Line downtime**	**32,021** blank cells (**88.95%**); **717** batches (**23.9%**) have 0 downtime	When we unpivot, Power Query removes blank cells, so those **717** zero-downtime batches disappear from **BatchDowntime**. To get the right *Average Downtime per Batch* (**30.72** mins instead of **40.37** mins), our DAX measure must divide by **COUNTROWS(ProductionBatch)**.
3	**Overnight Shifts Crossing Midnight**	**64** batches (**2.13%**) where **End Time < Start Time**	For batches that start before midnight and finish after midnight (e.g., **22:55** to **01:05**), subtracting **End Time - Start Time** gives a negative number. We will add **+ 1440** minutes (24 hours) whenever **End Time < Start Time**.
4	**Text Data Types for Times & Factor Headers**	**Start/End Time** stored as text; unpivoted **Factor** is text	We will convert **Start Time** and **End Time** to **Time**, and change the unpivoted **Factor** column from **Text** to **Whole Number** so it joins cleanly with **DowntimeFactor[Factor]**.
5	**No Separate Operator Table in Excel**	**4** operator names repeated across **3,000** rows	Our ERD has a separate **Operator** entity with **OperatorID** and **Name**. We will extract a distinct **Operator** table in Power Query and replace the text column with **OperatorID**.

3. Step-by-Step Data Cleaning Plan (In Power Query)

Since our entire project is implemented in **Power BI**, we will perform all data cleaning inside **Power Query Editor** before loading the tables into the Data Model:

1. **Import the 4 Data Tables & Clean Text:** Load **Line productivity**, **Line downtime**, **Products**, and **Downtime factors** into Power Query, and apply **Trim** and **Clean** on all text columns (**Product, Operator, Flavor, Size, Description**).
2. **Fix Date & Time Data Types:** Change **Date** from Date/Time to **Date** only, and set both **Start Time** and **End Time** to type **Time**.
3. **Create the Batch Duration (Mins) Column:** Add a Custom Column in **ProductionBatch** to calculate batch duration in minutes while fixing the 64 overnight shifts:

`if [End Time] < [Start Time] then Duration.TotalMinutes([End Time] - [Start Time]) + 1440 else Duration.TotalMinutes([End Time] - [Start Time])`

4. **Unpivot `Line downtime` into `BatchDowntime`:** Right-click the `Batch` column and select **Unpivot Other Columns**. Rename `Attribute` to `Factor` (change type to `Whole Number`) and `Value` to `DowntimeDuration` (change type to `Whole Number`). Add an Index Column named `DowntimeRecordID` starting from 1.
5. **Build the `Operator` Dimension Table:** Reference the `ProductionBatch` query, keep only the `Operator` column, click **Remove Duplicates**, add an Index Column (`OperatorID` from 1 to 4), and merge `OperatorID` back into `ProductionBatch` to match the ERD.
6. **Add Helper Columns & Date Table:** Merge `Min batch time` from `Product` (or use `RELATED()` in DAX) to check if `Batch Duration (Mins) > Min batch time`, and create a `Calendar` table in DAX for daily and monthly trend visuals.

4. ERD & Power BI Schema Design (Model View)

Our ERD (`Manifacturing Downtime ERD.pdf`) maps directly into a clean **Star / Snowflake Schema** inside Power BI's **Model View** using 5 tables connected by **One-to-Many (1 : *)** relationships:

- `Operator (1)` ⟶ `(*) ProductionBatch` on `OperatorID`: One operator handles many batches (Charlie: 892, Dee: 821, Mac: 693, Dennis: 594).
- `Product (1)` ⟶ `(*) ProductionBatch` on `ProductID`: One product is produced in many batches across the year.
- `ProductionBatch (1)` ⟶ `(*) BatchDowntime` on `Batch`: One batch can have zero, one, or multiple downtime occurrences.
- `DowntimeFactor (1)` ⟶ `(*) BatchDowntime` on `Factor`: One downtime factor is recorded across many downtime occurrences.

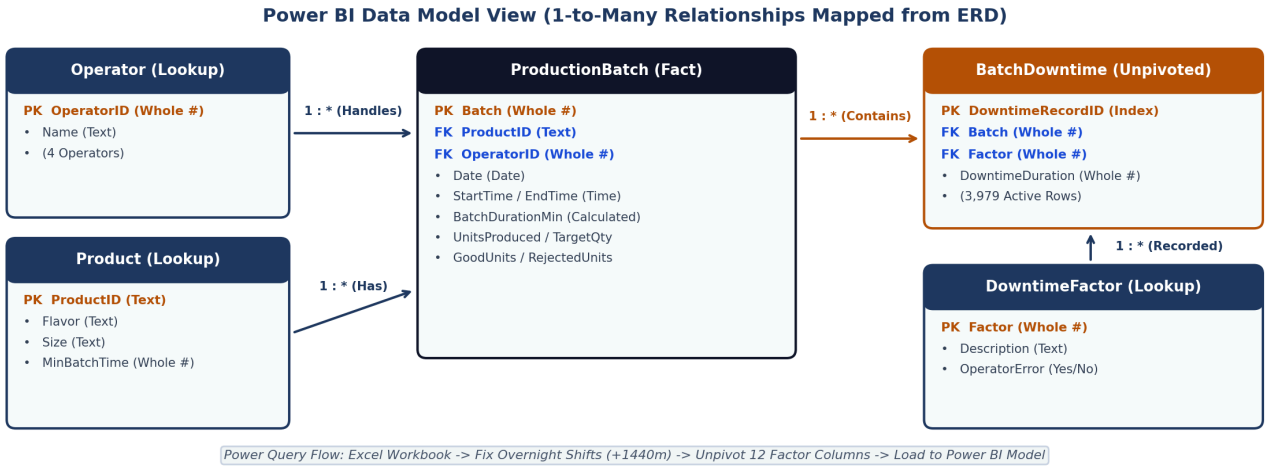


Figure 1: Power BI Model View Schema showing our 5 tables, Primary/Foreign Keys, 1-to-Many relationships, and Power Query flow.

Filter Propagation in Power BI: When a user selects a `Product`, `Operator`, or `Date` on the dashboard, the filter flows from the lookup tables into `ProductionBatch` and automatically filters `BatchDowntime`. To allow filtering batch counts by a specific `DowntimeFactor`, we can either enable bi-directional cross-filtering on the `BatchDowntime` relationship or use `CROSSFILTER()` inside that specific DAX measure.

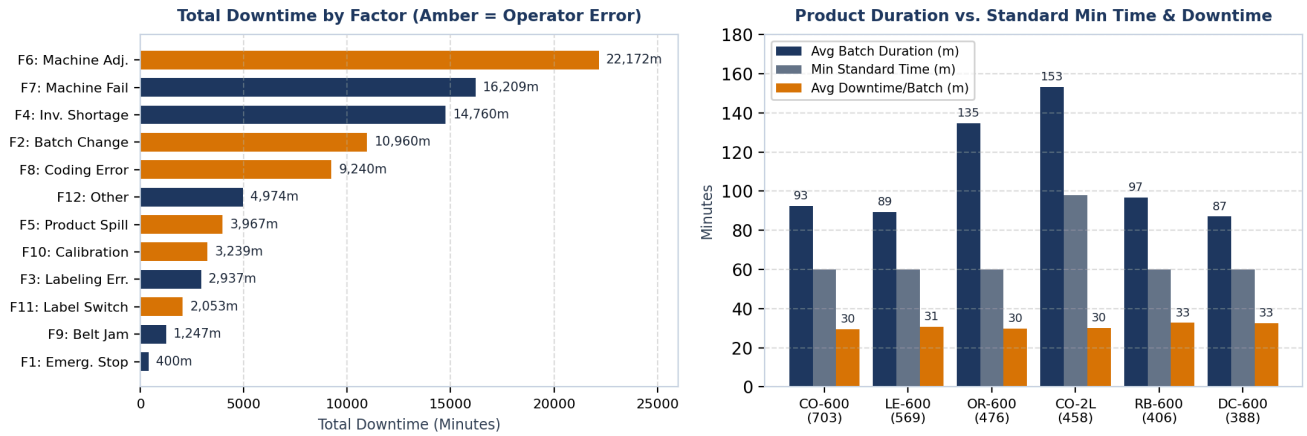


Figure 2: Initial Data Analysis — Left: Total downtime minutes by factor (orange = operator error); Right: Average batch duration vs. minimum standard time by product.

5. System Analysis & Design (DAX Measures & Dashboard Plan)

Our Power BI system follows a simple **4-layer workflow**: (1) **Excel Source** → (2) **Power Query Cleaning & Unpivoting** → (3) **Star/Snowflake Model & DAX Measures** → (4) **4-Page Interactive Report**. From our initial analysis shown in Figure 2, **56.02% of all downtime minutes** come from operator-related factors (led by *Machine adjustment* at 22,172 mins), and **95.27% of batches** exceed their minimum standard time. The table below outlines our core DAX measures and exact dataset benchmarks for Phase 3 testing:

Analysis Area	Power BI DAX Measures to Create	Exact Dataset Results (For Testing & Validation in Phase 3)
1. Downtime Analysis	Total Downtime = SUM(BatchDowntime[DowntimeDuration]) Avg Downtime = DIVIDE([Total Downtime], COUNTROWS(ProductionBatch))	**Total Downtime:** 92,158 mins (across 3,979 occurrences). **Avg Downtime per Batch:** 30.72 mins (across all 3,000 batches). **Batches with Downtime:** 2,283 batches (76.10%); 717 had zero downtime. **Top Factor (Time & Count):** Factor 6 - Machine adjustment (22,172 mins / 24.06% / 787 times), then Factor 7 - Machine failure (16,209 mins / 17.59% / 695 times).
2. Production Time Analysis	Total Duration = SUM(ProductionBatch[BatchDurationMin]) Avg Duration = AVERAGE(ProductionBatch[BatchDurationMin])	**Total Production Time:** 323,321 mins (5,388.7 hours). **Average Batch Duration:** 107.77 mins (Shortest: 60 mins, Longest: 219 mins). **Exceeded Min Batch Time:** 2,858 batches (95.27%) exceeded standard min time; 142 batches (4.73%) finished in the exact minimum time.

Analysis Area	Power BI DAX Measures to Create	Exact Dataset Results (For Testing & Validation in Phase 3)
3. Product & Operator Analysis	Total Batches = COUNTROWS(ProductionBatch) sliced by Product and Operator	**By Product:** Highest batches = **CO-600** (**703 batches**, **20,769** downtime mins), then **LE-600** (**569**), **OR-600** (**476**), **CO-2L** (**458**), **RB-600** (**406**), and lowest = **DC-600** (**388 batches**, **12,655** mins). **By Operator:** **Charlie** (**892 batches**, **692** with downtime, **27,419** mins), **Dee** (**821**, **613** with DT, **24,860** mins), **Mac** (**693**, **513** with DT, **20,477** mins), **Dennis** (**594**, **465** with DT, **19,402** mins).
4. Trends & Category Split	Breakdown by **Date**, **Size**, **Flavor**, and **Operator Error**	**By Bottle Size:** **600 ml** = **2,542 batches** (**84.73%**); **2 L** = **458 batches** (**15.27%**). **By Flavor:** **Cola** is #1 with **1,161 batches** (**38.70%**). **Operator Error Split:** Human error (**Yes**) caused **51,631 mins** (**56.02%**) vs. **40,527 mins** (**43.98%**) for machine/other factors (**No**).
5. Production Quantity & Target	Target Achievement % = DIVIDE(SUM([Units Produced]), SUM([Target Quantity])) Target Gap = [Total Produced] - [Total Target]	**Total Units Produced:** 4,185,326 units. **Total Target Quantity:** 4,198,389 units (**Target Gap:** -13,063 units). **Overall Target Achievement:** 99.69%. **Highest Product Achievement:** DC-600 (99.95%); **Lowest:** RB-600 (99.55%).
6. Quality & Performance	Good Rate % = DIVIDE(SUM([Good Units]), SUM([Units Produced])) Rejection Rate % = DIVIDE(SUM([Rejected Units]), SUM([Units Produced]))	**Total Good Units:** 4,089,359 (97.71%). **Total Rejected Units:** 95,967 (2.29%). **Best Operator Achievement:** Dennis (100.05%). **Lowest Operator Rejection Rate:** Dee (2.19%) vs. **Charlie** (2.38%).
7. Power BI Dashboard Pages	4 Interactive Report Pages with synced Slicers (**Date, Product, Operator, Factor**)	**Page 1 (Production Overview):** KPI cards (Produced, Target, Achievement %, Good Units, Downtime) + Daily production & downtime trends. **Page 2 (Downtime Analysis):** Pareto chart by factor, Operator Error donut, & Downtime vs. Output scatter chart. **Page 3 (Production Time Analysis):** Avg Duration vs. Min Batch Time & Actual vs. Target quantities by product. **Page 4 (Product & Operator Analysis):** Good vs. Rejected units, Rejection Rate %, and Operator scorecard table.